

Noah M Crabb

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EDUCATION

Virginia Polytechnic Institute, College of Engineering

Bachelor of Science: Aerospace and Ocean Engineering

Expected B.S Graduation: May 2027

GPA: 3.71/4.00 – *President's List Fall 2024, Dean's List Spring 2025*

Expected M.S Graduation: May 2028

Junior standing in an accelerated B.S./M.S. program (typically 5 years), on track to **complete both in 4 years**

Relevant Coursework: Astromechanics, Aerothermodynamics & Propulsion, Fluid Dynamics, Aerospace Structures, System Dynamics and Control, Computational Methods, Aerodynamics & Compressibility

WORK EXPERIENCE

Pi Sigma Epsilon – Delta Epsilon Chapter, Virginia Tech

Blacksburg, Virginia

Professional Member

Nov 2025 – Present

- Selected as **1 of 26 members from 120+ applicants** through a competitive, multi-week recruitment process including interviews, case studies, and professional evaluations.
- Completed an **intensive 8-week development program** with 4 meetings per week, focusing on communication, teamwork, project coordination, and business acumen—including **proficiency with Microsoft/Excel tools**.
- Synthesized **quantitative performance and financial results** into professional presentations, bridging engineering-style analysis with non-technical audiences.

3DPAC

Blacksburg, Virginia

CAD Lead / CAD Engineer

Jan 2025 – Present

- Promoted to CAD Lead in Mar. 2026** after serving as one of two CAD engineers responsible for full-aircraft CAD development; now oversee CAD members, delegate modeling tasks, track deadlines, lead Fall CAD recruitment, and coordinate design standards across the full team.
- Lead technical CAD meetings, member training, and formal design reviews** to teach Fusion360/SOLIDWORKS workflows, evaluate aircraft assemblies, resolve design and manufacturability issues, and keep competition deliverables on schedule.
- Co-developed complete competition aircraft CAD models and supported rapid design–build–test cycles using **Creativity 3D printers**, contributing to a national **1st Place Most Innovative Design** award and **2nd Place Time Aloft** finish at UT Arlington's 3D Printed Aircraft Competition.

PROJECTS

Wind Tunnel Boundary Layer & Sting Mount Research Project

Blacksburg, Virginia

Experimental Aerodynamics Team Member

Apr 2026

- Conducted wind tunnel testing in the **Virginia Tech Stability Wind Tunnel**, measuring turbulent boundary layer velocity profiles with a 30-port pneumatic rake across multiple Reynolds number conditions
- Processed pressure data in **MATLAB** to compute normalized velocity profiles, local freestream velocity, and boundary layer thickness, comparing experimental results to turbulent boundary layer power-law predictions.
- Analyzed the effect of **sting mount geometry** on near-wall flow development, connecting boundary layer growth trends to adverse pressure gradients, horseshoe vortex behavior, aerodynamic blockage, and CFD validation.

Machine Learning Assisted Turbofan Engine Performance Optimizer

Blacksburg, Virginia

Independent Project

June 2026

- Developed a **MATLAB based turbofan performance optimizer** using thermodynamic cycle analysis and machine learning surrogate modeling, generating 8,000 simulated engine-cycle samples to predict thrust, fuel consumption, and efficiency trends.
- Built an interactive **engineering GUI** that enables users to adjust key turbofan design variables, run baseline cycle calculations, train AI models, and compare **thermal, propulsive, and overall efficiency** outputs in one tool.
- Implemented **optimization and sensitivity analysis features** to identify the most influential performance drivers across the design space, improving evaluation speed and supporting faster **AI-assisted propulsion trade studies**.

OTHER

- Skills:** SOLIDWORKS, Fusion360, Python, MATLAB, Excel, Revit, XFMR5, Microsoft 365, AutoCAD, aircraft assembly design, aerodynamic analysis, boundary layer analysis, wind tunnel testing, 3D printing, rapid prototyping.
- Interests:** Pickleball, Soccer, CAD-Based Design Challenges, Plane spotting, Propulsion